Introduction to AI

CS 151: Introduction to Cybersecurity

Emily Hand

A little about me

Associate Professor in Computer Science and Engineering

Director of the Machine Perception Lab

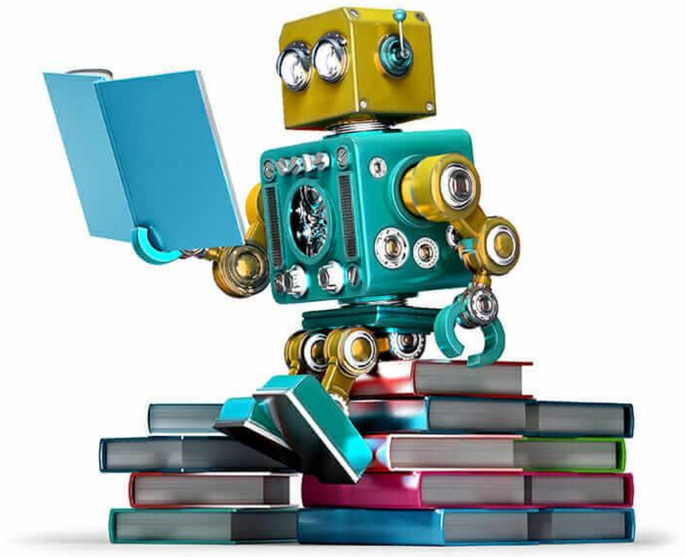
Member of the Computer Vision Lab

Affiliate Faculty with

Cybersecurity Center

Integrative Neuroscience Program

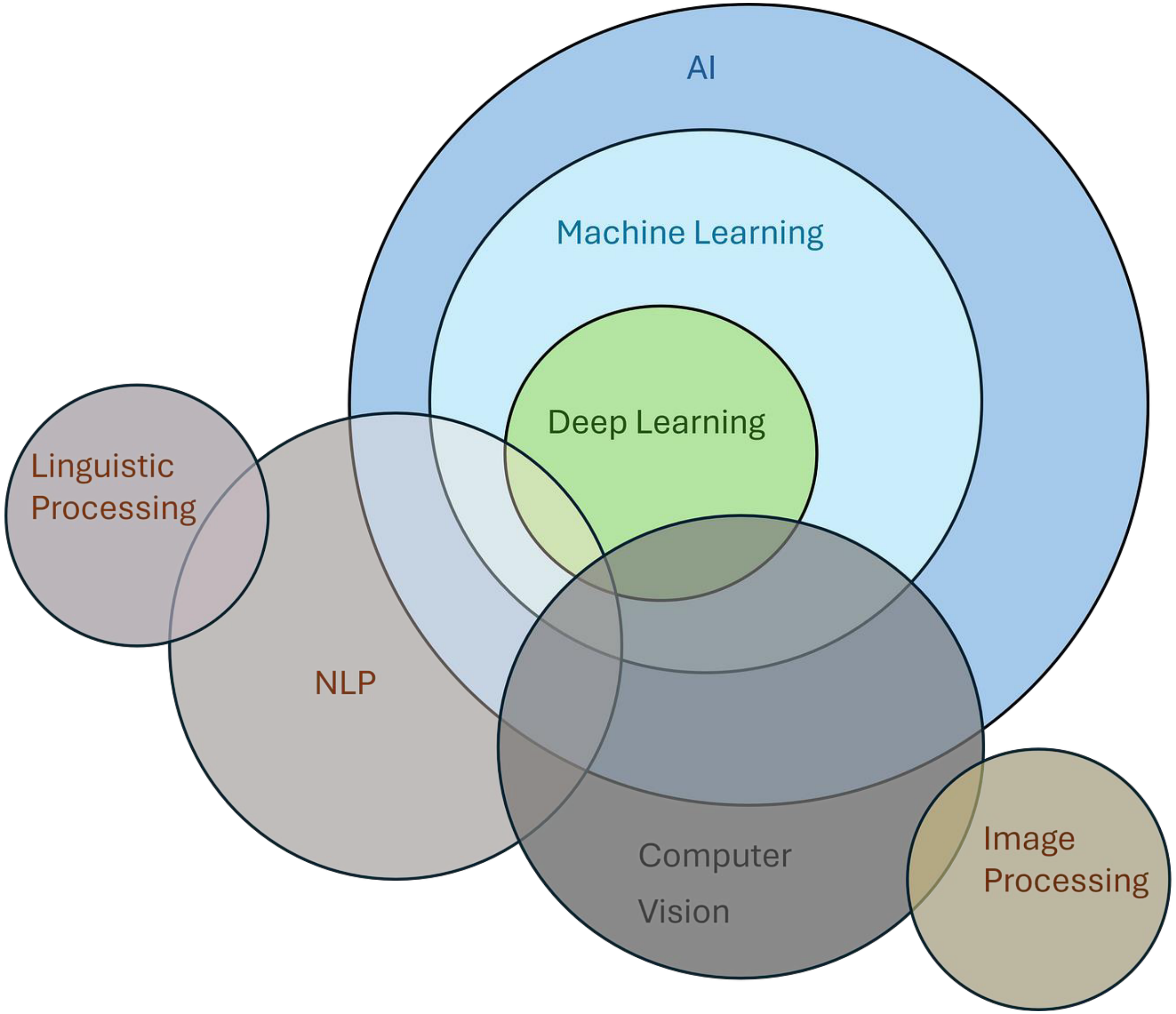
Artificial Intelligence



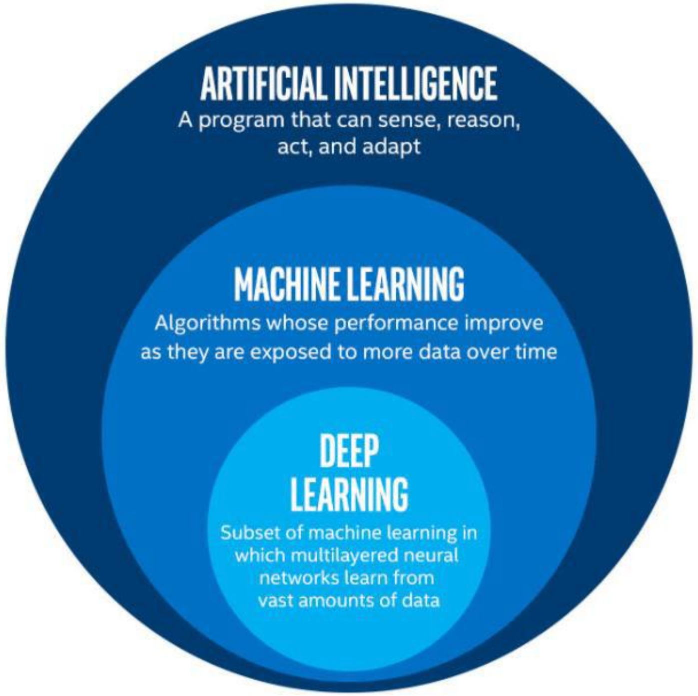
Intelligence

the ability to acquire and apply knowledge and skills

AI


AI
Machine Learning
Deep Learning
Linguistic Processing
NLP
Computer Vision
Image Processing

AI


ARTIFICIAL INTELLIGENCE
A program that can sense, reason, act, and adapt
MACHINE LEARNING
Algorithms whose performance improve as they are exposed to more data over time
DEEP LEARNING
Subset of machine learning in which multilayered neural networks learn from vast amounts of data

AI vs. ML

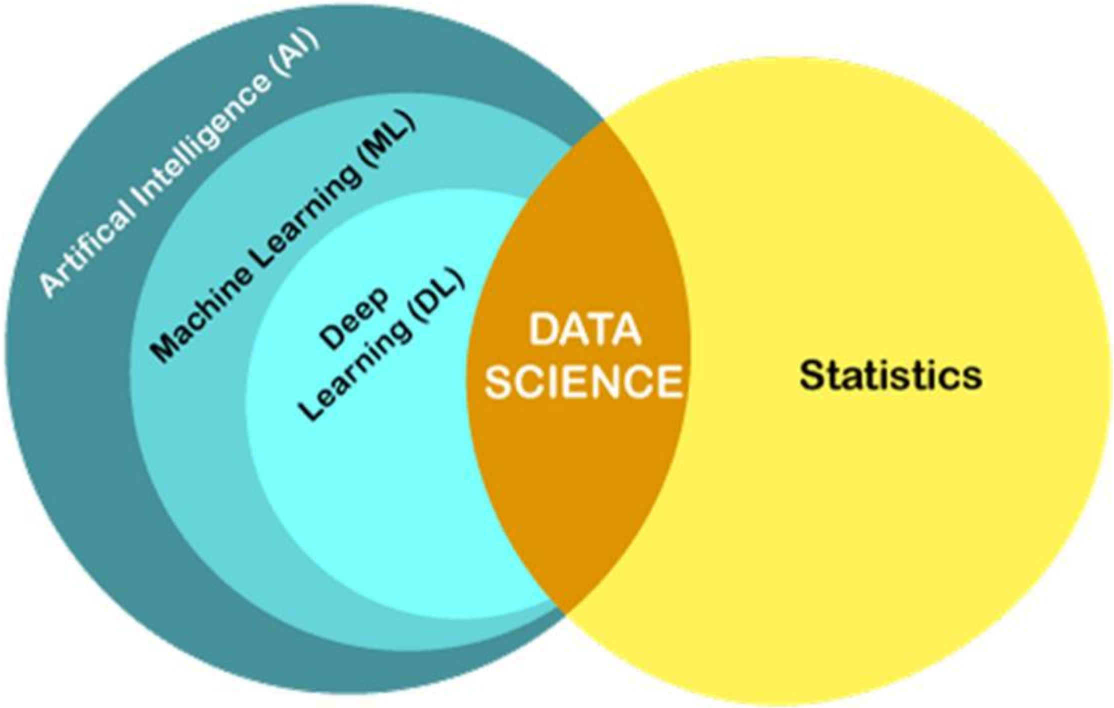
AI

Expert Systems

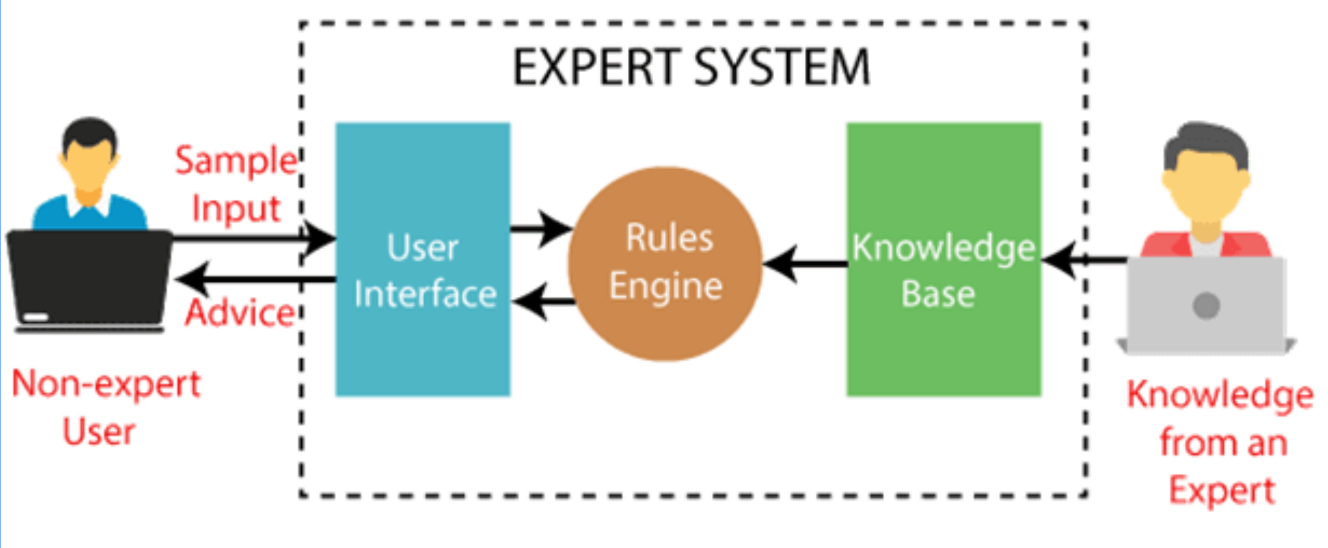
ML

Improves Performance on a Task given Experience

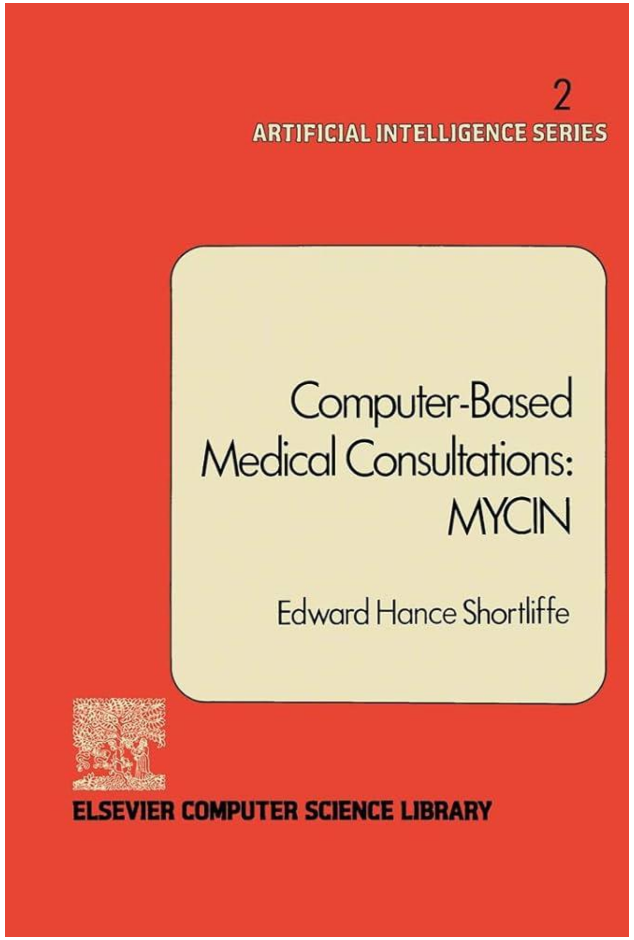
AI vs. ML


Artifical Intelligence (AI)
Machine Learning (ML)
Deep Learning (DL)
DATA SCIENCE
Statistics

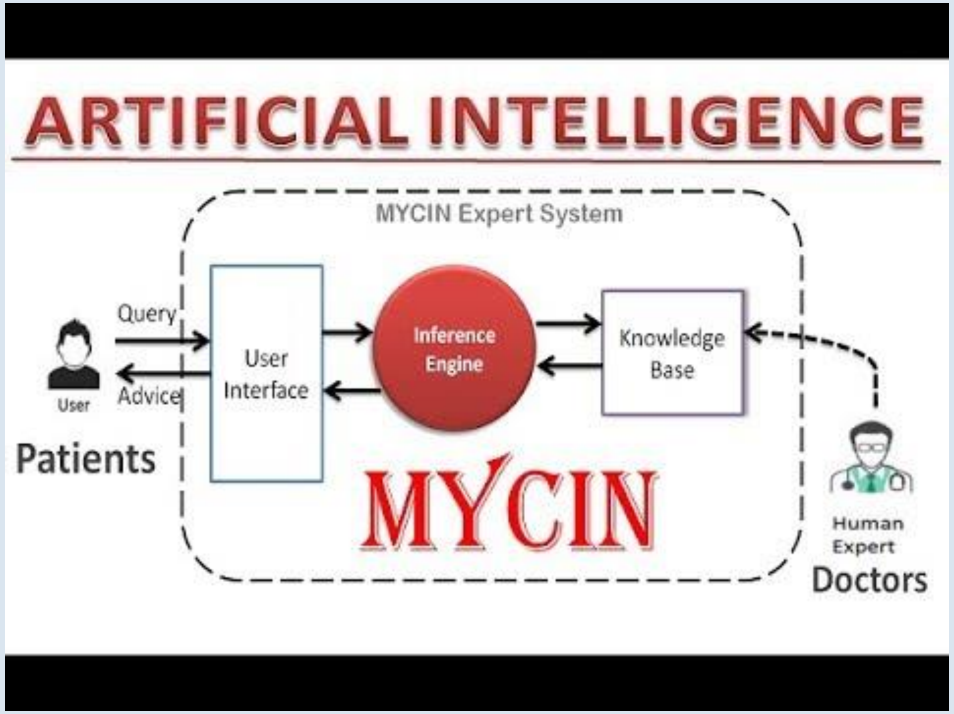
AI: Expert Systems



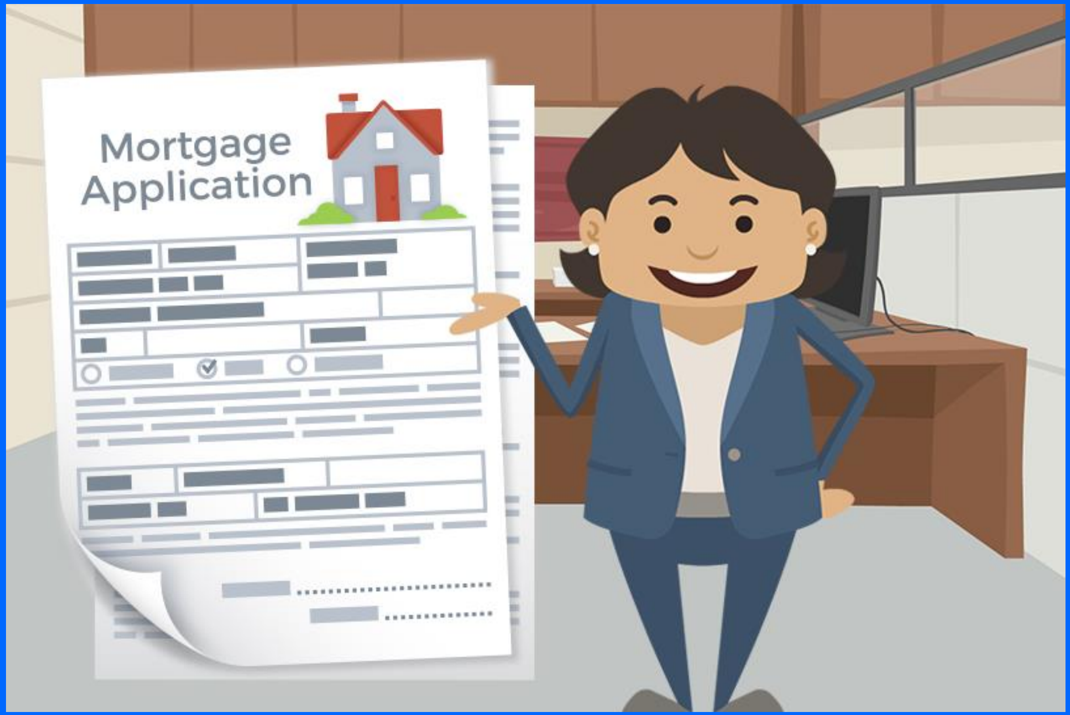
AI: Expert Systems



AI: Expert Systems



AI: Expert Systems



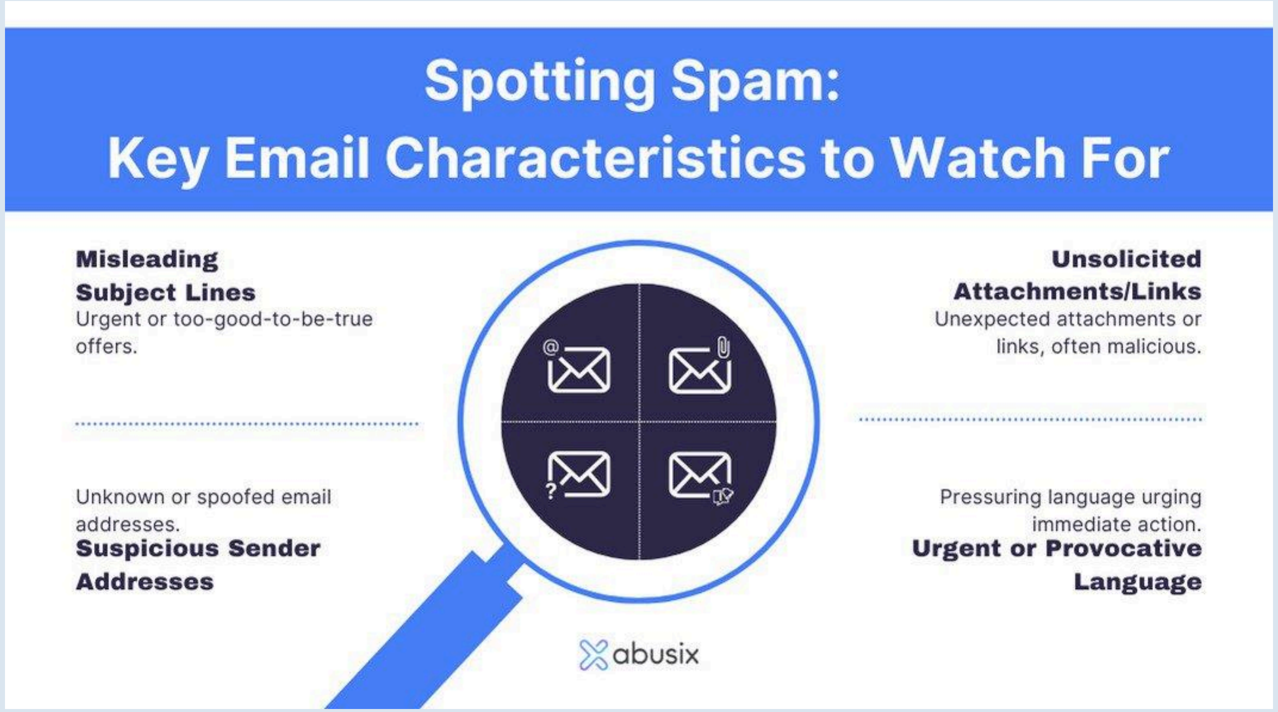
AI: Expert Systems

AI: Expert Systems

VitalLaw™

(formerly known as Cheetah)

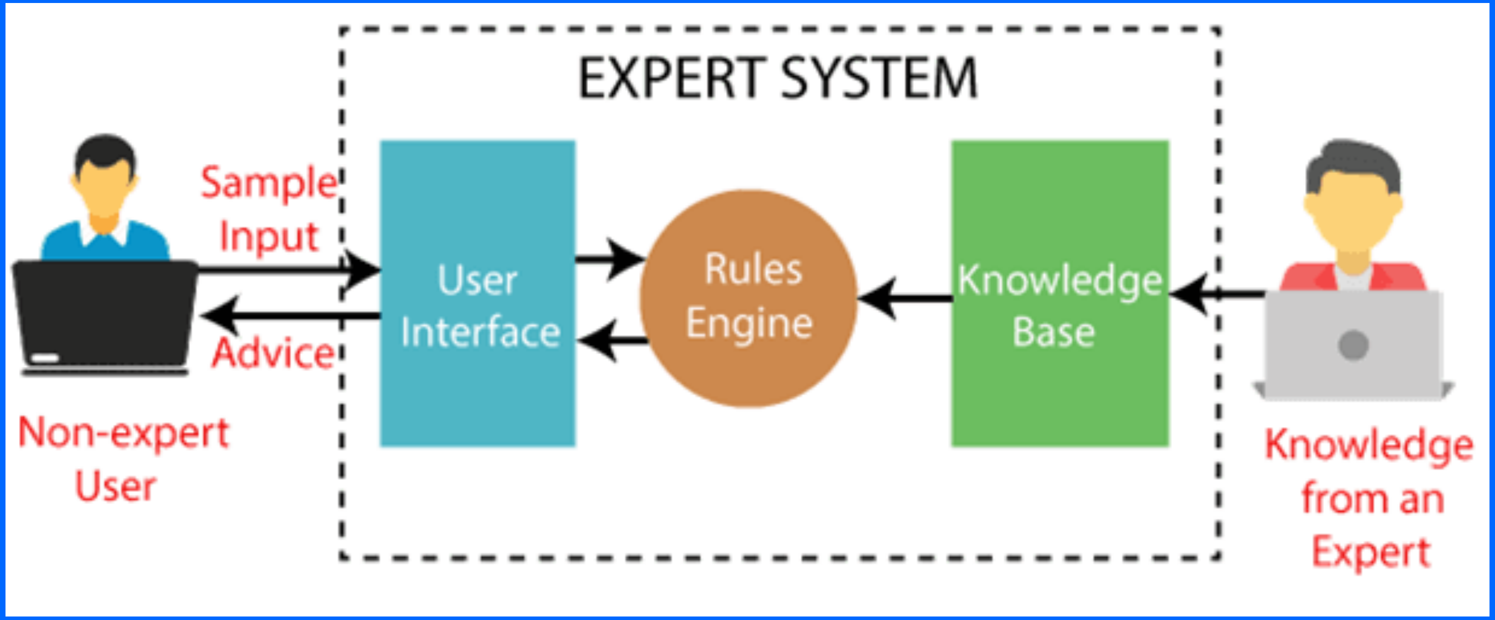
AI: Expert Systems



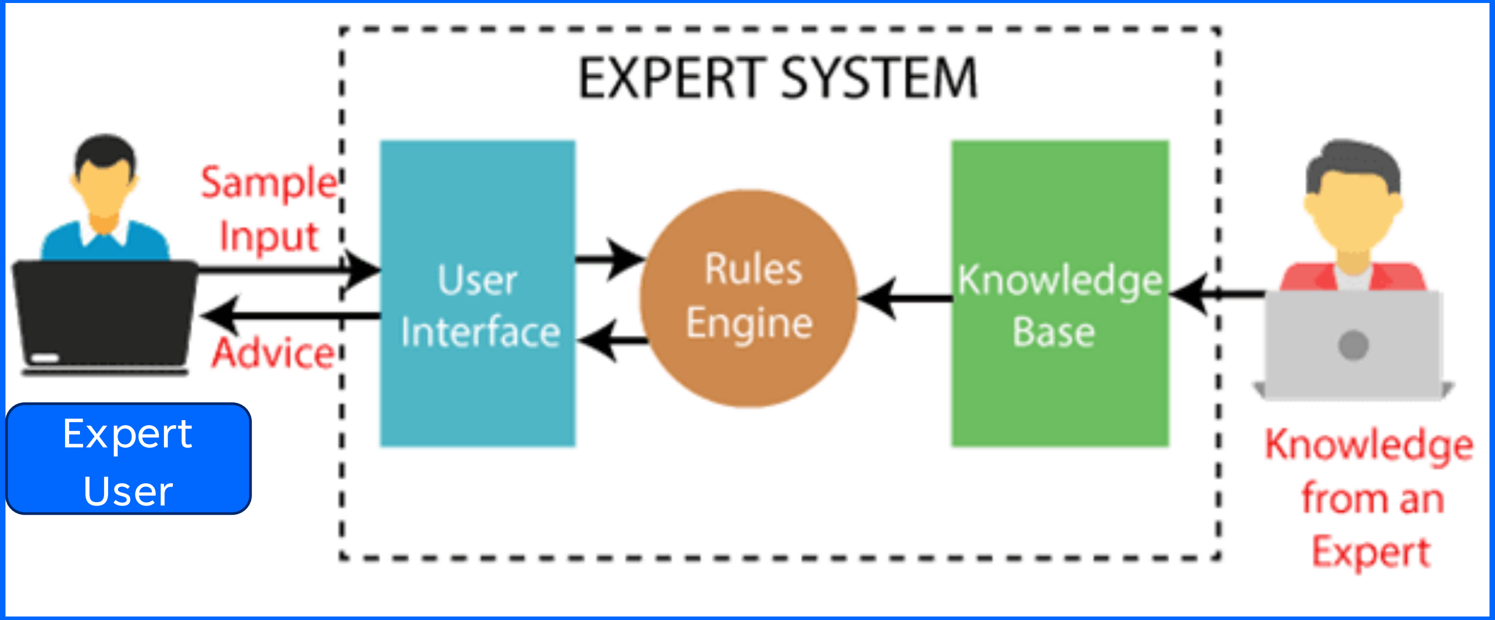
Machine Learning

Improving Performance on a Task Given Experience

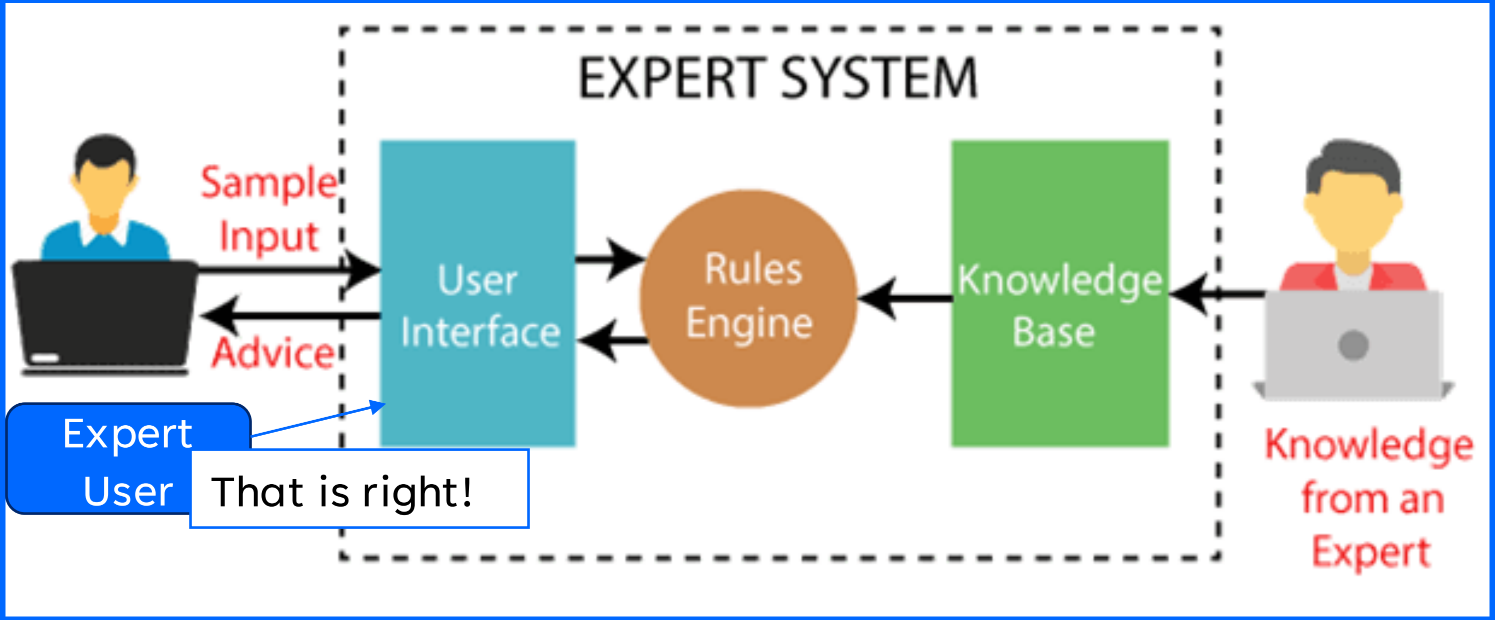
Machine Learning



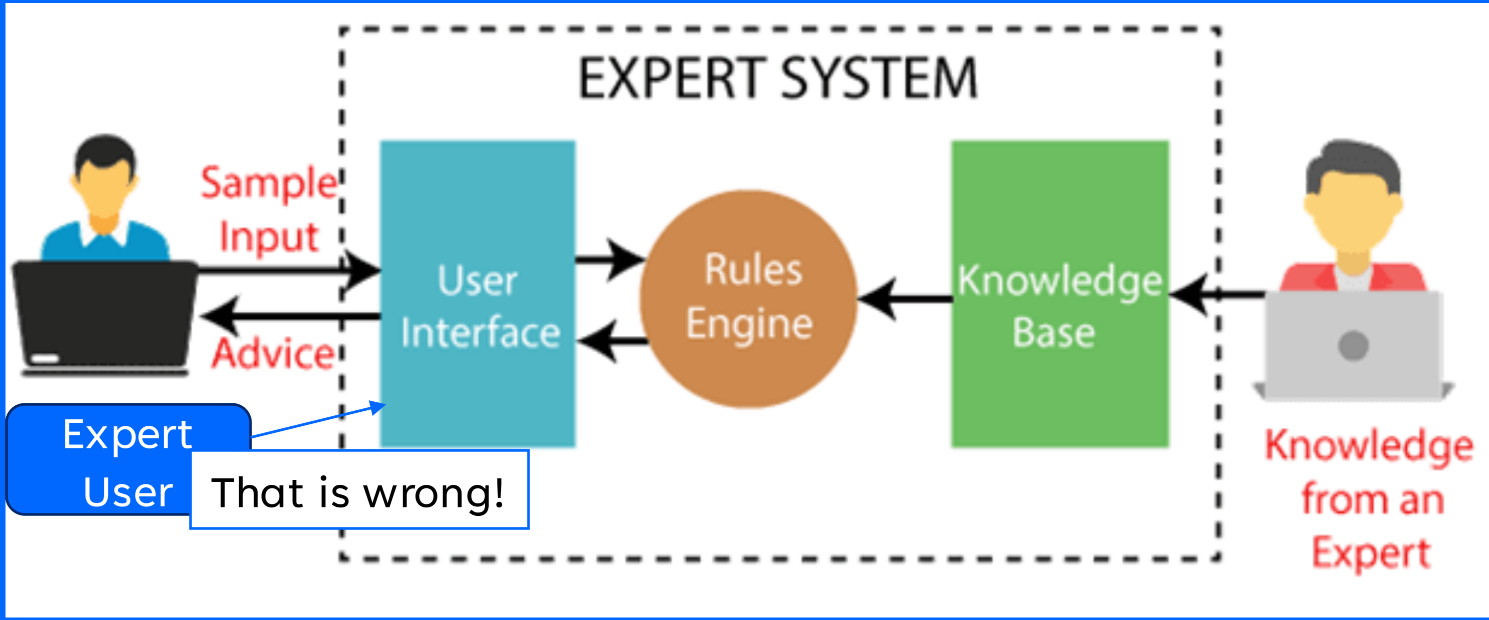
Machine Learning



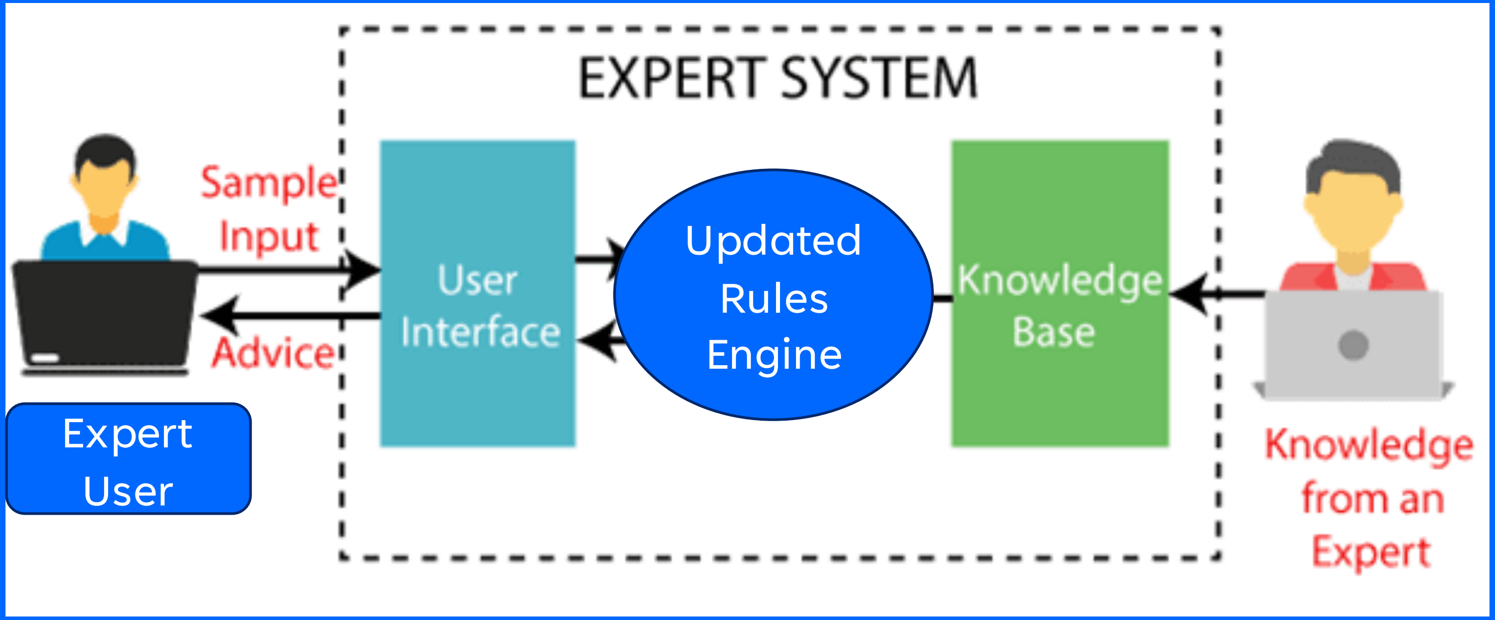
Machine Learning



Machine Learning



Machine Learning



Machine Learning

Algorithm that aims to improve

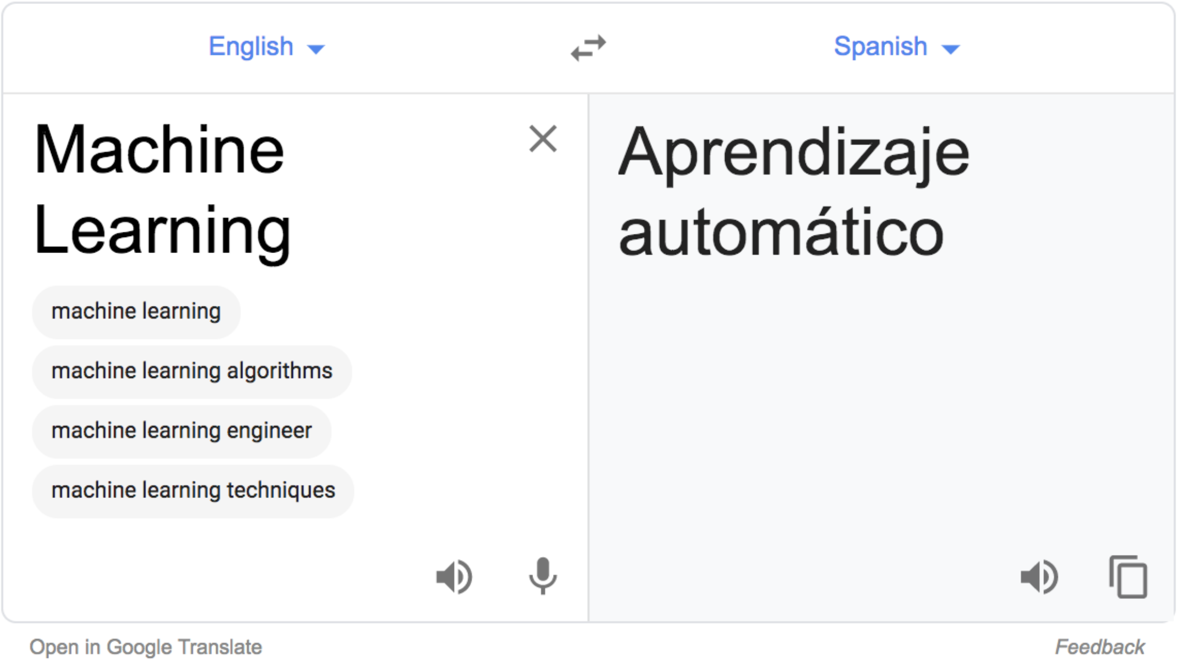
- **P**erformance on a
- **T**ask with
- **E**xperience

All of these must be specified.

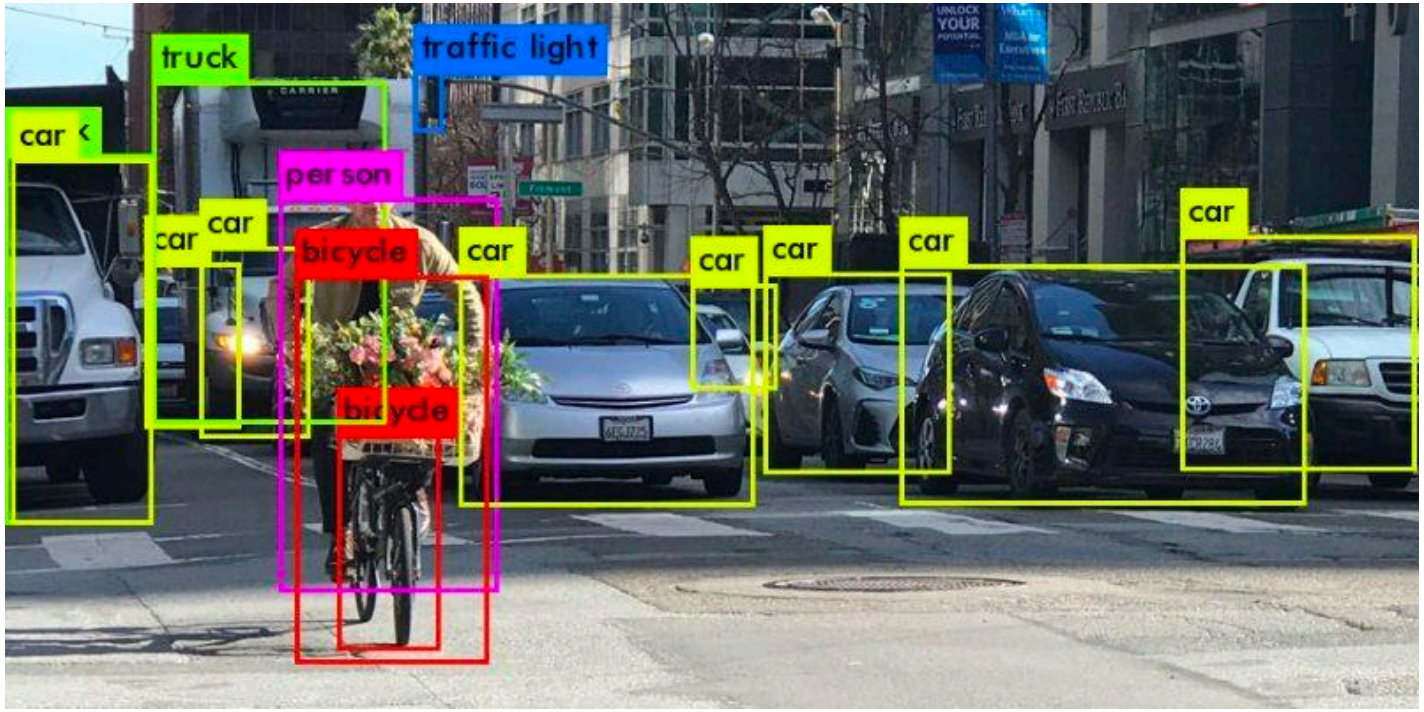
- Performance: Machine Translation vs. Object Detection
- Task: What is important for learning?
- Experience: What information does the algorithm have access to?

Generative AI

Translation



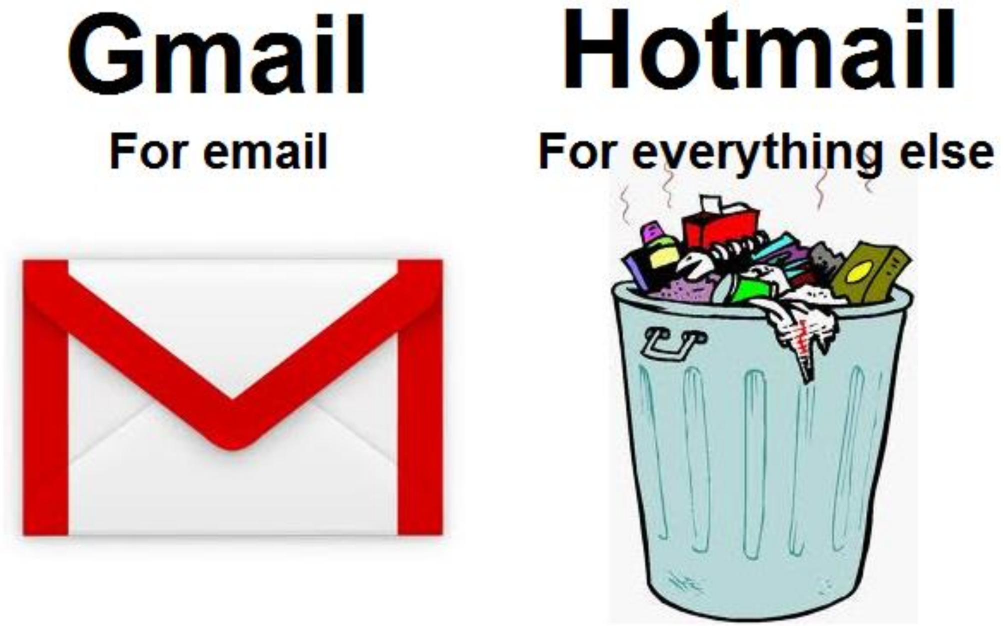
Object Detection



Face Recognition

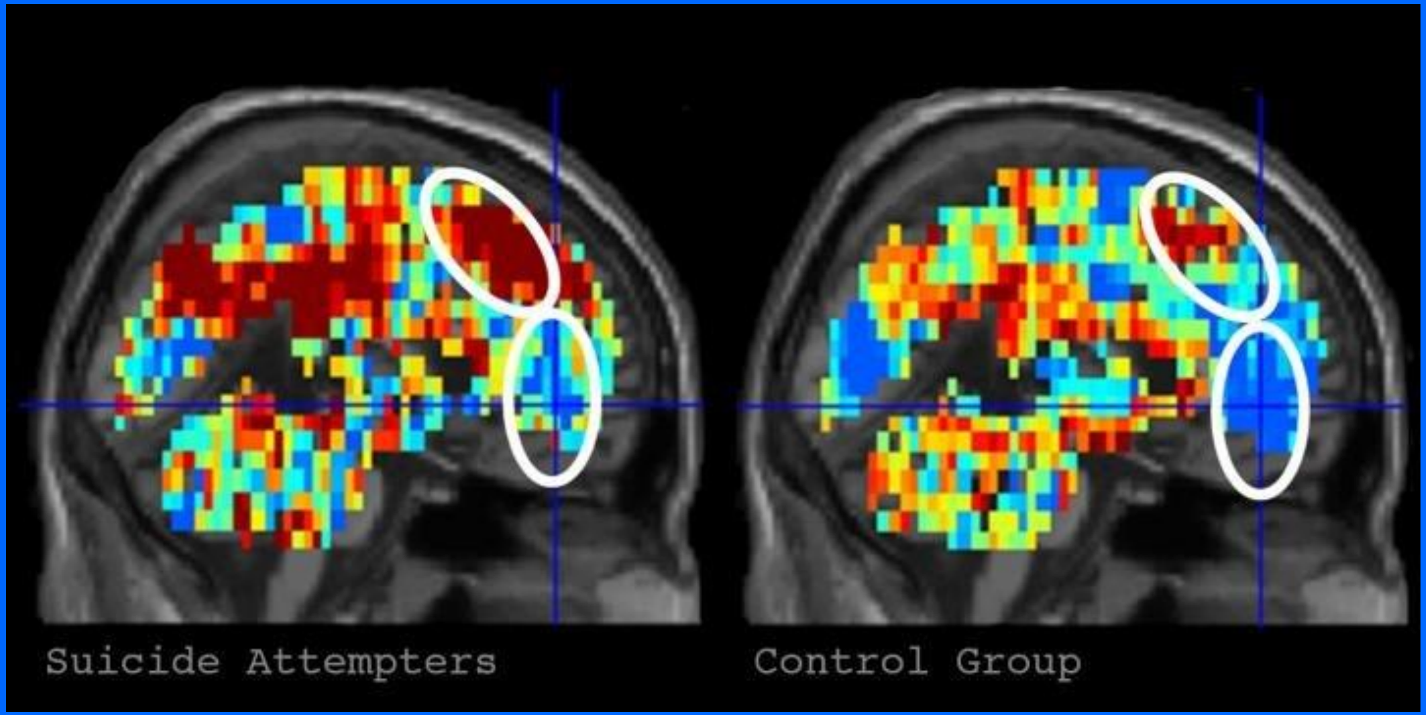
Attribute Recognition

Spam Detection

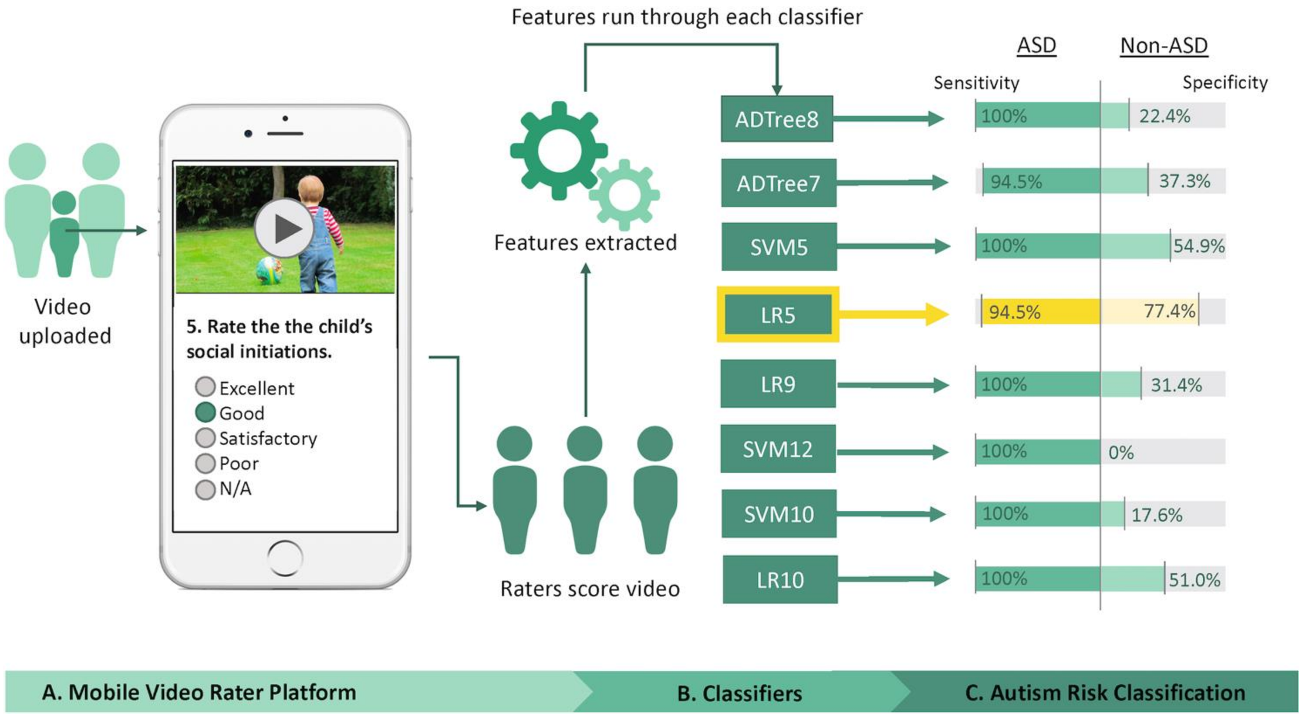


Custom Predictions

Suicide Prevention



Autism Recognition



Limitations

Limitations

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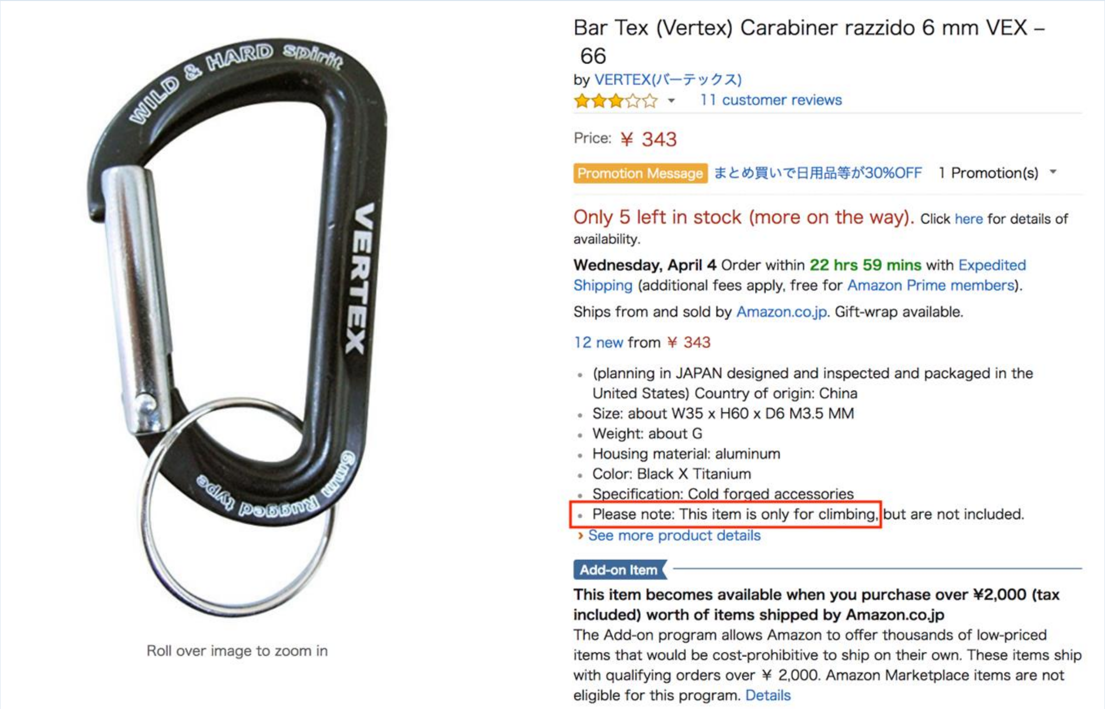


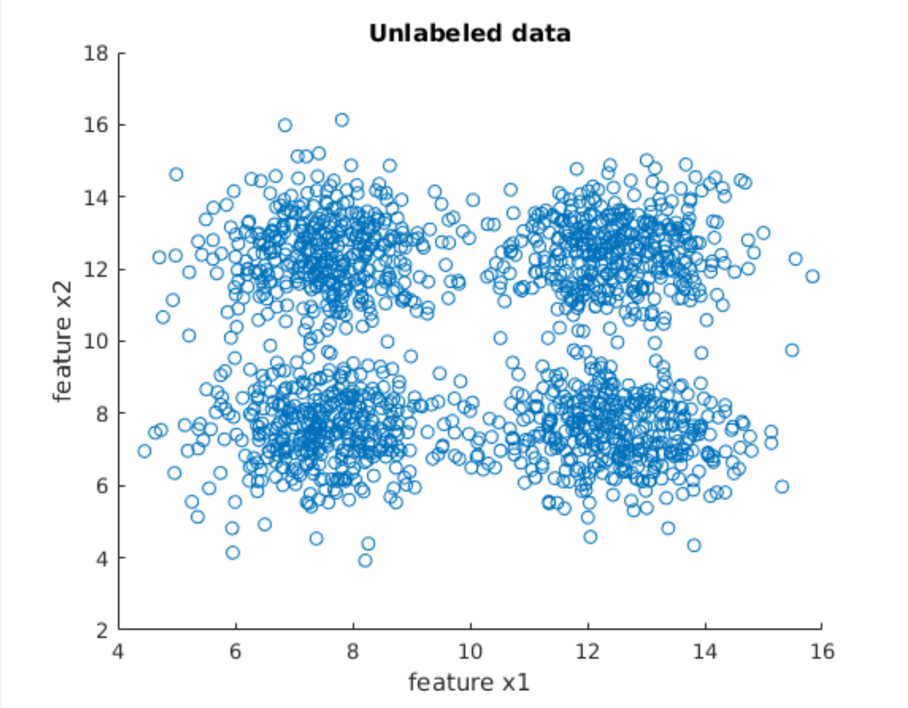
Limitations

ML Problem

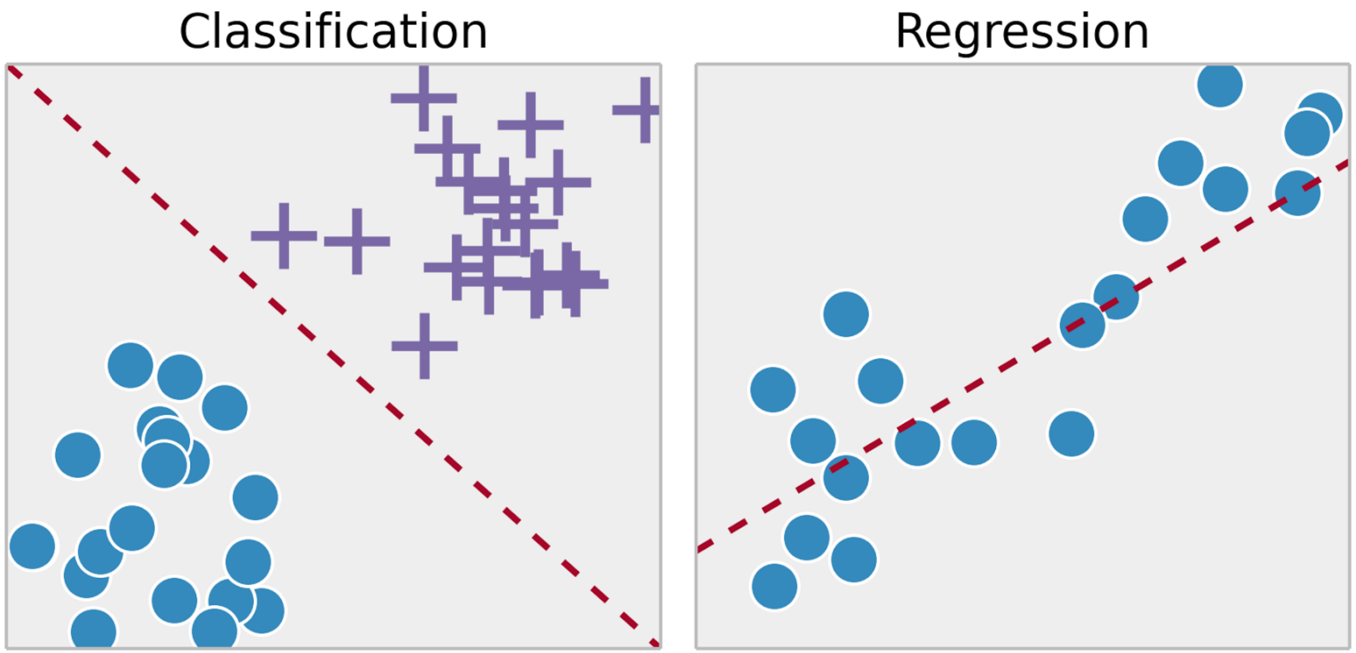
Features

Labels

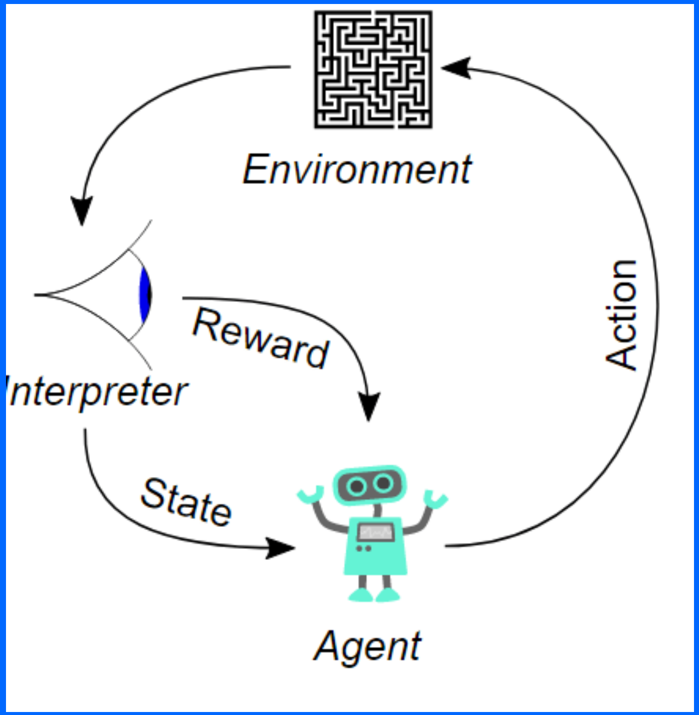
Unsupervised Learning



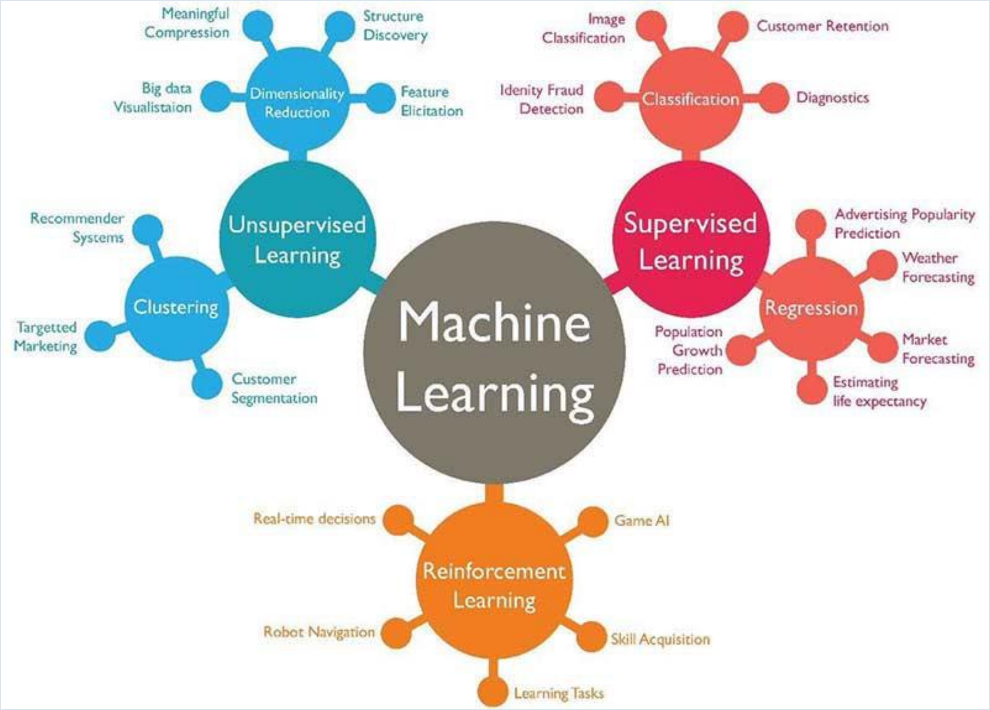
Supervised Learning



Reinforcement Learning



Machine Learning

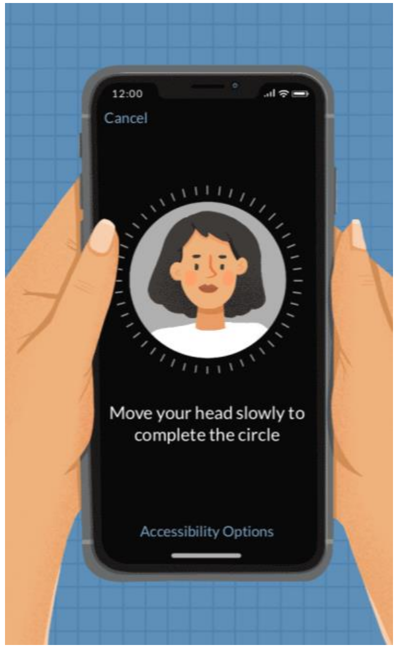


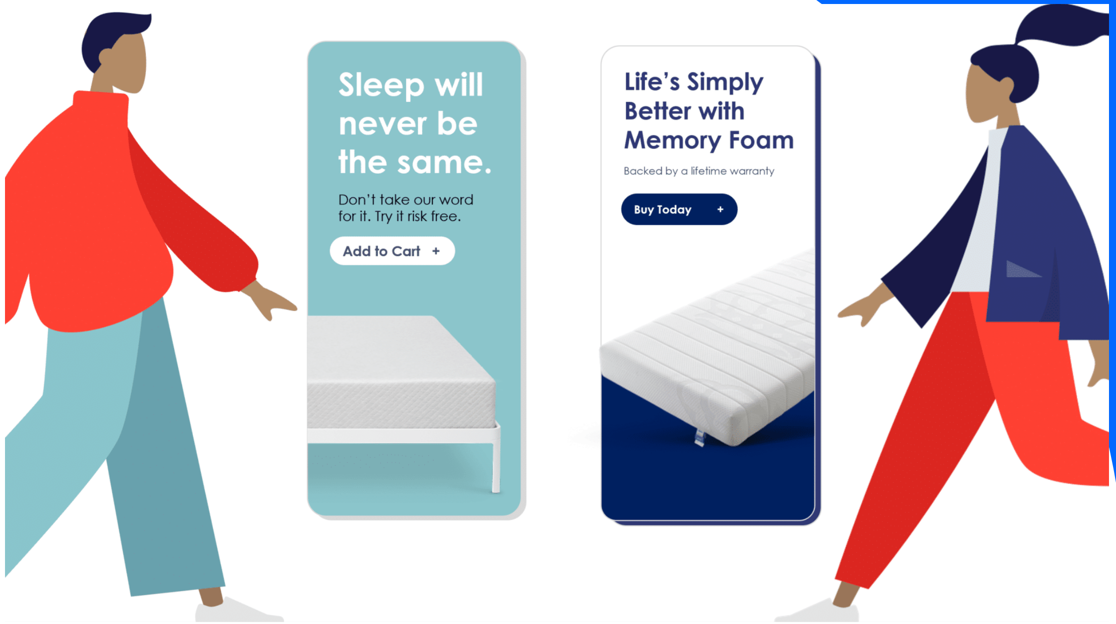
Uses & Misuses of AI

How is AI used today?

Uses


amazon


12:00
Cancel
Move your head slowly to complete the circle
Accessibility Options


Sleep will never be the same.
Don't take our word for it. Try it risk free.
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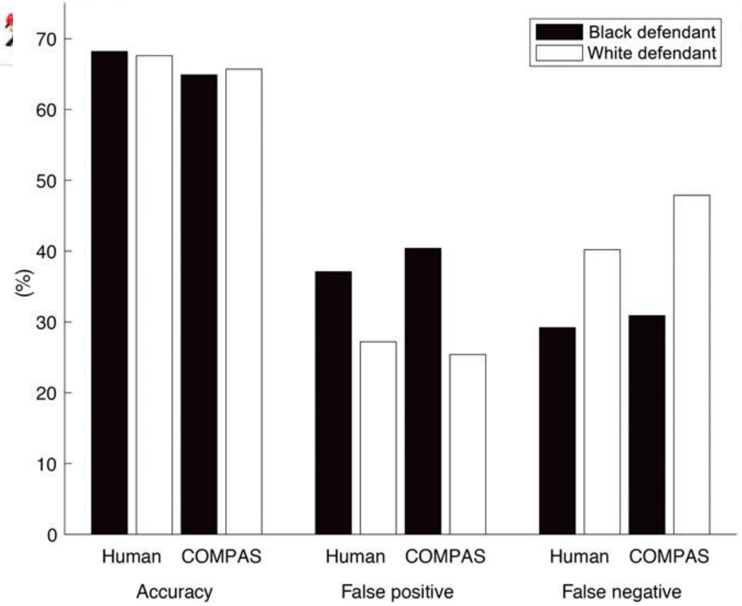
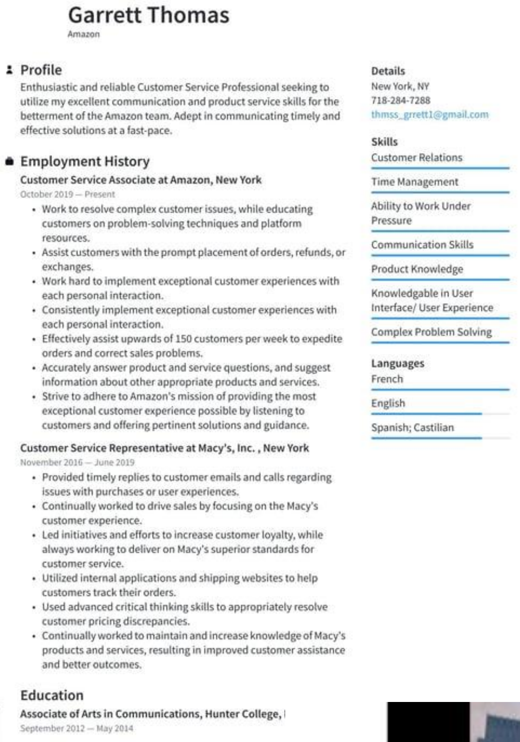
Backed by a lifetime warranty
Buy Today +


EMAIL FILTER
inbox
Spam

Misuses





Garrett Thomas

Amazon

Profile

Enthusiastic and reliable Customer Service Professional seeking to utilize my excellent communication and product service skills for the betterment of the Amazon team. Adept in communicating timely and effective solutions at a fast-pace.

Employment History

Customer Service Associate at Amazon, New York

October 2019 — Present

- Work to resolve complex customer issues, while educating customers on problem-solving techniques and platform resources.
- Assist customers with the prompt placement of orders, refunds, or exchanges.
- Work hard to implement exceptional customer experiences with each personal interaction.
- Consistently implement exceptional customer experiences with each personal interaction.
- Effectively assist upwards of 150 customers per week to expedite orders and correct sales problems.
- Accurately answer product and service questions, and suggest information about other appropriate products and services.
- Strive to adhere to Amazon's mission of providing the most exceptional customer experience possible by listening to customers and offering pertinent solutions and guidance.

Customer Service Representative at Macy's, Inc. , New York

November 2016 — June 2019

- Provided timely replies to customer emails and calls regarding issues with purchases or user experiences.
- Continually worked to drive sales by focusing on the Macy's customer experience.
- Led initiatives and efforts to increase customer loyalty, while always working to deliver on Macy's superior standards for customer service.
- Utilized internal applications and shipping websites to help customers track their orders.
- Used advanced critical thinking skills to appropriately resolve customer pricing discrepancies.
- Continually worked to maintain and increase knowledge of Macy's products and services, resulting in improved customer assistance and better outcomes.

Education

Associate of Arts in Communications, Hunter College,

September 2012 — May 2014

Details

New York, NY

718-284-7288

thmss_grett1@gmail.com

Skills

Customer Relations

Time Management

Ability to Work Under Pressure

Communication Skills

Product Knowledge

Knowledgable in User Interface/ User Experience

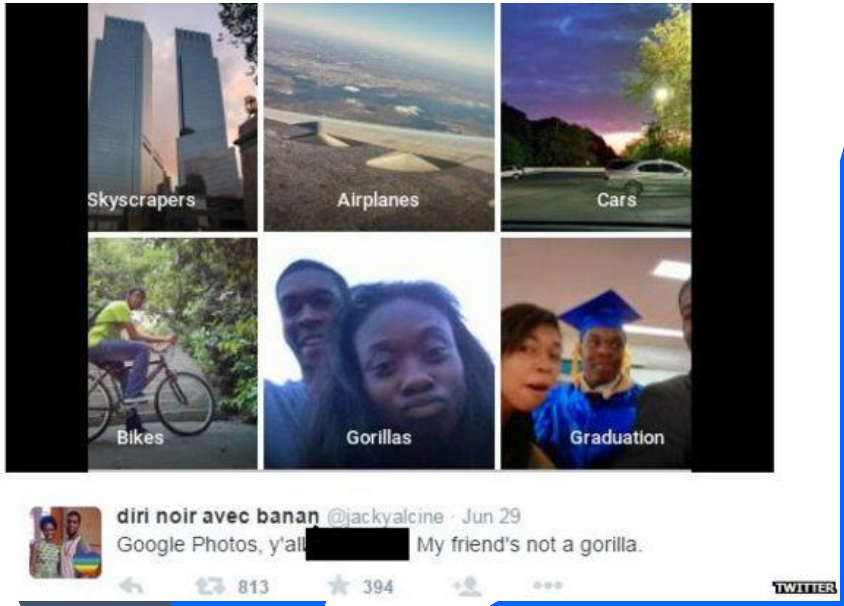
Complex Problem Solving

Languages

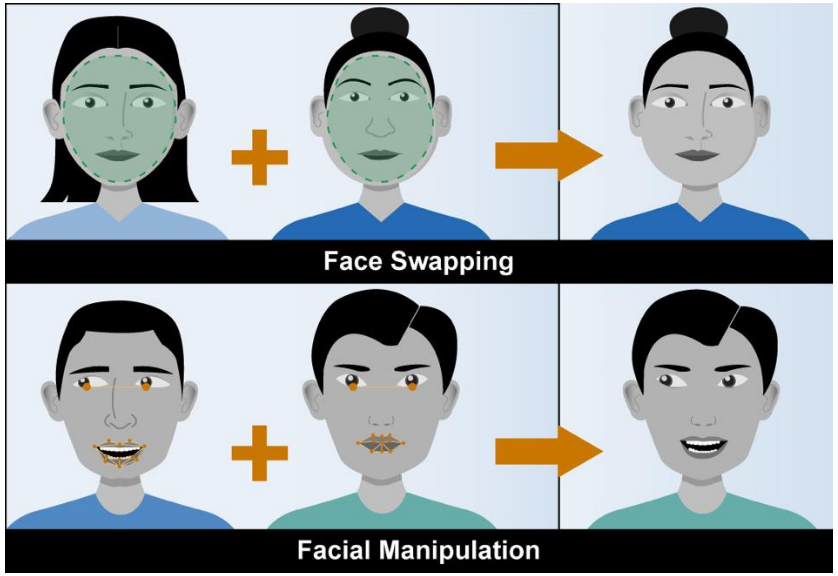
French

English

Spanish; Castilian



Misuses



Uses and Misuses of AI

Why do we have misuses?

What leads to misuses?

- Young Field
- Competition
- Code Availability

Young Field

Both CS and AI

Engineering

Computer Science

Software Engineering

Hardware Engineering

Computer Engineering

Computer Science

Programming

Computer Science vs. Engineering

Much younger discipline

100s of years old vs. 1000s

Artificial Intelligence

1950s

Recent History of AI

Rule-Based

Statistics

Pattern Recognition

Machine Learning

Deep Learning

Competition

Driving Innovation

Innovation

Universities

Companies

Universities

Publish or perish model

Hire multiple people for one position

Research lifecycle became too fast paced

From journals to conferences

Arxiv

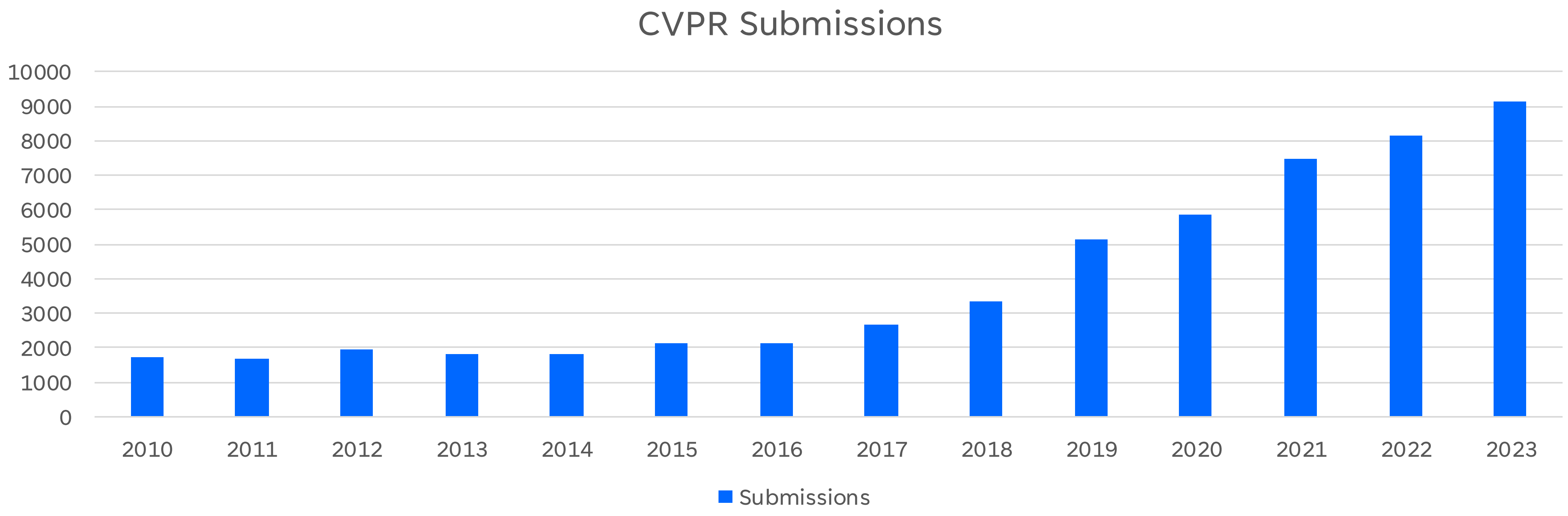
Companies

Used to take research from universities and build on it

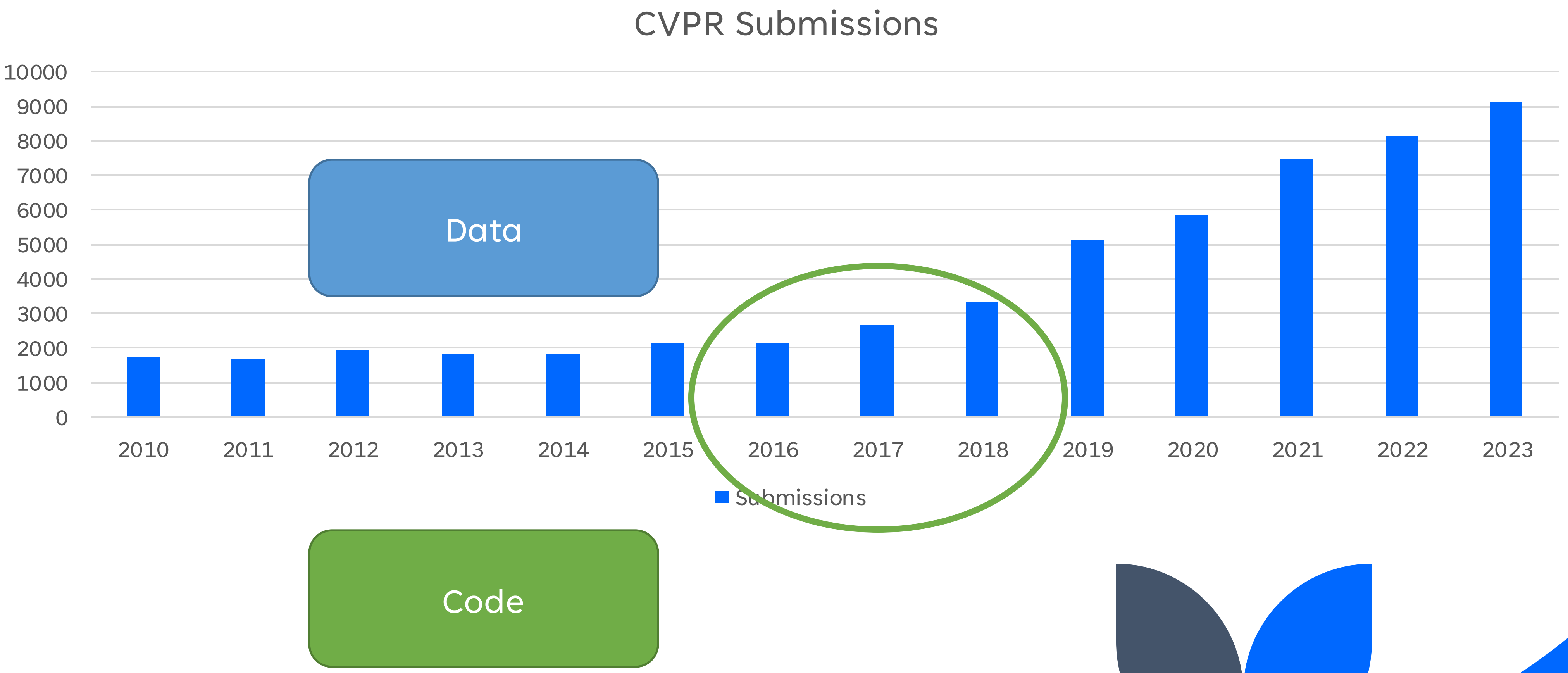
Now in direct competition with universities

Have much larger research arms

Oversaturated



What happened?



Code Availability

Submitting code with publications

Code Availability

Code standards have changed

Everyone makes their code/demos available

Oftentimes before publication

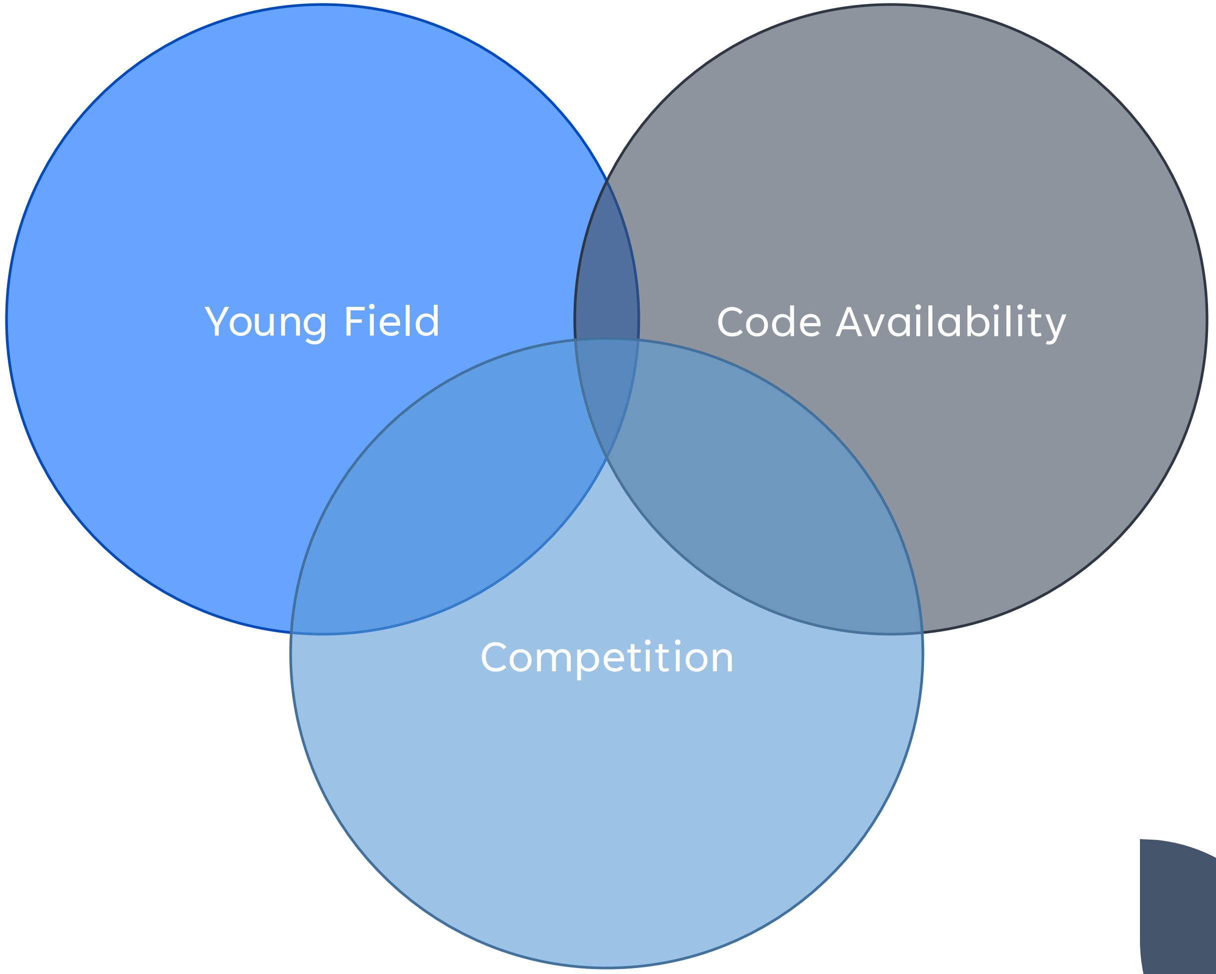
Reproducibility isn't considered important anymore

Anyone can download and run

What leads to misuses?

- Young Field
- Competition
- Code Availability

Perfect Storm



So, what is being done?

Explainable AI (XAI)

Limitations and Ethical Considerations in Publications

Workshops/Conferences on Ethical AI

Introducing bias/fairness/transparency/robustness metrics

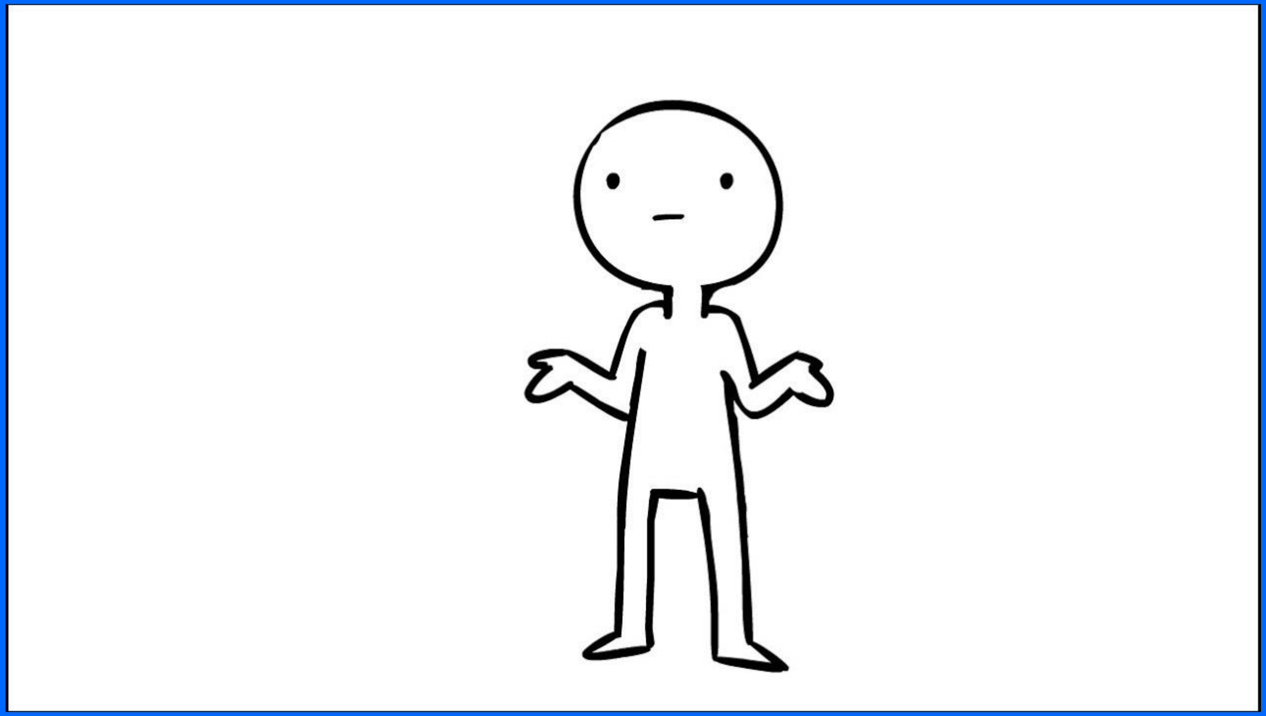
What’s next?

Regulations?

Collaborations with Philosophers and Ethicists?

Moratorium on innovation?

What’s next?



Thank you

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www.machineperceptionlab.com